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United States Patent	12400769
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Ma; Donghui et al.

Residual stress improvement method and residual stress improvement apparatus

Abstract

A method of improving residual stress in a canister in which fuel assemblies are loaded includes, when the canister having a cylindrical canister shell and a canister lid welded to the upper opening of the canister shell is disposed in a cylindrical cask body, filling the space above and around an upper end portion of the canister in the cask body with fill water. Then, compressive stress is applied to a predetermined region of the upper end portion of the canister in the fill water. This makes it possible to readily apply compressive stress to the canister while shielding radioactive rays by the fill water.

Inventors:	Ma; Donghui (Osaka, JP), Okada; Jun (Osaka, JP), Goto; Masanori (Osaka, JP)
Applicant:	KANADEVIA CORPORATION (Osaka, JP)
Family ID:	1000008781039
Assignee:	KANADEVIA CORPORATION (Osaka, JP)
Appl. No.:	17/927283
Filed (or PCT Filed):	March 03, 2021
PCT No.:	PCT/JP2021/008115
PCT Pub. No.:	WO2021/240938
PCT Pub. Date:	December 02, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230207147 A1	Jun. 29, 2023

Foreign Application Priority Data

JP 2020-092325 May. 27, 2020

Publication Classification

Int. Cl.: G21F5/12 (20060101); G21F5/008 (20060101); G21F5/06 (20060101)

U.S. Cl.:

CPC G21F5/12 (20130101); G21F5/008 (20130101); G21F5/06 (20130101);

Field of Classification Search

CPC: G21F (5/008); G21F (5/06); G21F (5/12); G21F (9/36); B23K (2101/12); C21D (7/06); B21D (31/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4197467	12/1979	Williams	976/DIG.344	G21F 5/10
5553106	12/1995	Enomoto	134/108	G21C 13/087
9508459	12/2015	Kitagawa	N/A	G21F 9/36
11250962	12/2021	Singh	N/A	G21F 5/008
2011/0005288	12/2010	Matsui et al.	N/A	N/A
2016/0240275	12/2015	Kitagawa et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
6-79626	12/1993	JP	N/A
07-328860	12/1994	JP	N/A
2001-343490	12/2000	JP	N/A
2002-162391	12/2001	JP	N/A
2003-255082	12/2002	JP	N/A
5208064	12/2012	JP	N/A
2014-181978	12/2013	JP	N/A
6208962	12/2016	JP	N/A

OTHER PUBLICATIONS

CN 103521919A, Luo et al. Jan. 2014. cited by examiner
CN 109580342A, Peng et al. Apr. 2019. cited by examiner
WO 97/39457A1, Jung et al. Oct. 1997. cited by examiner
Search Report dated Apr. 27, 2021, issued in corresponding International Application No. PCT/JP2021/008115 with English translation (4 pgs.). cited by applicant
Written Opinion of the International Searching Authority dated Apr. 27, 2021, issued in corresponding International Application No. PCT/JP2021/008115 (4 pgs.). cited by applicant

Primary Examiner: Tolan; Edward T

Attorney, Agent or Firm: Pillsbury Winthrop Shaw Pittman, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. National Stage of PCT/JP2021/008115, filed Mar. 3, 2021, which claims priority to Japanese Patent Application No. 2020-092325, filed May 27, 2020, the entire contents of both of which are herein incorporated by reference in their entireties.

TECHNICAL FIELD

(2) The present invention relates to a technique for improving residual stress in a canister in which fuel assemblies are loaded.

CROSS-REFERENCE TO RELATED APPLICATIONS

(3) This application claims priority benefit of Japanese Patent Application No. JP2020-092325 filed in the Japan Patent Office on May 27, 2020, the entire disclosure of which is incorporated herein by reference.

BACKGROUND ART

(4) Spent fuel assemblies taken out of a nuclear reactor (hereinafter, simply referred to as “fuel assemblies”) are temporarily stored in a storage pool in a building and then loaded in a storage canister. The canister that is housed in a transport cask is transported to a storage facility and rehoused in, for example, a concrete storage container.

(5) Japanese Patent No. 6208962 (Document 1) discloses a technique for preventing stress-corrosion cracking in a canister. With this technique, residual tensile stress that occurs is cancelled by applying first compressive stress in advance to an area of a cylindrical shell (canister shell) in which residual tensile stress is expected to occur due to welding of a lid (canister lid) and, with residual compressive stress applied to this area, welding the canister lid to the canister shell. Thereafter, second compressive stress is applied to only the upper end of the canister shell and an axial area in the vicinity of the upper end, both of which become close to a melting state during welding.

(6) Document 1 also discloses a technique for inserting the canister into a transport cask (cask body) and subjecting the canister shell to plastic working with use of an upper opening (ring-shaped gap) formed between the transport cask and the canister, and a technique for filling the upper opening with water and cooling welds with water during the welding of the canister lid so as to narrow the axial area in which residual tensile stress occurs after welding. Japanese Patent No. 5208064 (Document 2) describes a method of water-jet peening performed on a reactor core internal structure installed in a nuclear reactor pressure container.

(7) In the case where compressive stress is applied to a canister that is disposed in a cylindrical cask body as in Document 1, it is not possible to shield radioactive rays coming from the inside of the canister toward the upper opening of the cask body. This increases the difficulty of the operation of applying compressive stress to the canister.

SUMMARY OF THE INVENTION

(8) The present invention is intended for a residual stress improvement method of improving residual stress in a canister in which fuel assemblies are loaded, and it is an object of the present

invention to readily apply compressive stress to the canister.

(9) A preferable residual stress improvement method according to the present invention includes a) when a canister having a cylindrical canister shell and a canister lid welded to an upper opening of the canister shell is disposed in a cylindrical cask body, filling a space above and around an upper end portion of the canister in the cask body with fill water, and b) applying compressive stress to a predetermined region of the upper end portion of the canister in the fill water.

(10) According to the present invention, it is possible to readily apply compressive stress to the canister while shielding radioactive rays by the fill water.

(11) Preferably, in the operation b), water-jet peening is used to apply compressive stress to the predetermined region, the water-jet peening being performed by injecting water from a nozzle that is disposed in the fill water.

(12) Preferably, an annular space is formed between an inner peripheral surface of the cask body and an outer peripheral surface of the canister and filled with water, and in the operation b), the water injected from the nozzle includes the water that has filled the annular space.

(13) Preferably, the fill water in the operation a) and the water injected from the nozzle in the operation b) are clean water that contains no radioactive materials.

(14) Preferably, in the operation a), a surface of water is formed in a ring-shaped weir that is attached to an upper portion of the cask body and projects upward therefrom.

(15) Preferably, an annular space is formed between an inner peripheral surface of the cask body and an outer peripheral surface of the canister, and a width of the annular space around the upper end portion of the canister is greater than a width of the annular space around a central portion of the canister, and when the canister lid is welded to the upper opening of the canister shell, a ring-shaped shield is placed around the upper end portion of the canister.

(16) The present invention is also intended for a residual stress improvement apparatus for improving residual stress in a canister in which fuel assemblies are loaded. A preferable residual stress improvement apparatus includes a water filler that, when a canister having a cylindrical canister shell and a canister lid welded to an upper opening of the canister shell is disposed in a cylindrical cask body, fills a space above and around an upper end portion of the canister in the cask body with fill water, and a compressive stress applicator that applies compressive stress to a predetermined region of the upper end portion of the canister in the fill water.

(17) These and other objects, features, aspects and advantages of the present invention will become more apparent from the following detailed description of the present invention when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating a configuration of a residual stress improvement apparatus;

(2) FIG. 2 is a diagram illustrating an area in the vicinity of the upper end portion of a canister;

(3) FIG. 3 is a flowchart of processing relating to the loading of fuel assemblies into the canister;

(4) FIG. 4 is a diagram for describing the processing relating to the loading of fuel assemblies into the canister;

(5) FIG. 5 is a diagram for describing the processing relating to the loading of fuel assemblies into the canister;

(6) FIG. 6 is a diagram for describing the processing relating to the loading of fuel assemblies into the canister;

(7) FIG. 7 is a diagram for describing the processing relating to the loading of fuel assemblies into the canister; and

(8) FIG. 8 is a diagram for describing the processing relating to the loading of fuel assemblies into

the canister.

DESCRIPTION OF EMBODIMENTS

(9) FIG. 1 is a diagram illustrating a configuration of a residual stress improvement apparatus 1 according to one embodiment of the present invention. FIG. 1 shows a cross section including the central axes of a cask 7 and a canister 8, which will be described later (the same applies to the other drawings that illustrate the cask 7 and the canister 8). The residual stress improvement apparatus 1 is an apparatus for improving residual stress in the canister 8 in which fuel assemblies 9 are loaded. For example, the residual stress improvement apparatus 1 may be provided on an operation floor F (or in a decontamination pit or the like) located adjacent to a storage pool P in a nuclear reactor building. The storage pool P stores water, and the fuel assemblies 9 are immersed and stored in this water. The water in the storage pool P (hereinafter, referred to as “pooled water”) is polluted water that contains radioactive materials.

(10) The residual stress improvement apparatus 1 includes a cask holder 11, a ring-shaped weir 12, a water filler 2, and a compressive stress applicator 3. On the cask holder 11, a cask body 71 of the transport cask 7 is placed. The cask 7 may be formed of a neutron shielding material such as resin and metal, or may be formed of concrete or the like. The cask body 71 may be placed directly on the operation floor F. The cask body 71 has a cylindrical shape having a central axis pointing in the up-down direction in FIG. 1 as its center. In the example illustrated in FIG. 1, the cask body 71 has an approximately cylindrical shape. The cask body 71 may have any other shape such as a polygonal cylindrical shape. The upper and lower portions of the cask body 71 are provided respectively with an upper water hole 711 and a lower water hole 712. The upper water hole 711 and the lower water hole 712 penetrate through the side wall of the cask body 71. The upper water hole 711 and the lower water hole 712 allow the supply of water into the cask body 71 and the collection of water from the cask body 71.

(11) The cask body 71 has a lower opening provided with an approximately disk-like cask bottom 72. The lower opening of the cask body 71 is closed by the cask bottom 72. The cask bottom 72 may be detachable from the cask body 71 and fixedly attached by, for example, bolting to the cask body 71. Depending on the design of the cask 7, the cask bottom 72 may be formed integrally with the cask body 71. In the state illustrated in FIG. 1, the cask body 71 has an upper opening that is open. The upper opening of the cask body 71 is closed by the cask lid 73 that is fixedly attached by, for example, bolting to the upper opening through operations described later (see FIG. 8). The cask 7 primarily includes the cask body 71, the cask bottom 72, and the cask lid 73.

(12) The canister 8 is disposed in the cask body 71. For example, the canister 8 may be formed of metal such as stainless steel, and a plurality of fuel assemblies 9 may be loaded in the canister 8. The canister 8 includes a canister shell 81, a canister bottom 82, and a canister lid 83. The canister shell 81 has a cylindrical shape having a central axis pointing in the up-down direction in FIG. 1 as its center. In the example illustrated in FIG. 1, the canister shell 81 has an approximately cylindrical shape. The canister shell 81 may have any other shape such as a polygonal cylindrical shape. The canister shell 81 has a lower opening provided with an approximately disk-like canister bottom 82. The lower opening of the canister shell 81 is closed by the canister bottom 82.

(13) Typically, the canister bottom 82 is welded and fixedly attached to the lower opening of the canister shell 81 before the fuel assemblies 9 are loaded in the canister shell 81. Before the fuel assemblies 9 are loaded in the canister shell 81, the canister shell 81 is also subjected to plastic working (the application of compressive stress) by shot peening or any other technique in order to improve, in advance, residual stress that is caused by the welding of the canister bottom 82. Here, the improvement of the residual stress refers to reducing residual tensile stress that is caused by welding (including the case of shifting the residual stress to the compressive side; the same applies below). Depending on the design, the canister bottom 82 may be formed integrally with the canister shell 81.

(14) The canister shell 81 has an upper opening provided with an approximately disk-like canister

lid **83**. The upper opening of the canister shell **81** is closed by the canister lid **83**. As will be described later, the canister lid **83** is welded and fixedly attached to the upper opening of the canister shell **81** after the fuel assemblies **9** are loaded in the canister shell **81**. Besides, residual stress that is caused by the welding of the canister lid **83** is improved by processing of the residual stress improvement apparatus **1** described later. In the following description, the canister lid **83** and a portion of the canister shell **81** that is located in the vicinity of the canister lid **83** are collectively referred to as an “upper end portion **86**.” The upper end portion **86** includes a portion of the canister **8** in which residual tensile stress occurs when the canister lid **83** is welded to the upper opening of the canister shell **81**.

(15) In the example illustrated in FIG. **1**, the canister **8** is disposed in the cask body **71** so that the central axis of the cylindrical cask body **71** overlaps approximately with the central axis of the cylindrical canister shell **81**. The inside diameter of the cask body **71** is slightly greater than the outside diameter of the canister shell **81**. Thus, an annular space **79** is formed between the inner peripheral surface of the cask body **71** and the outer peripheral surface of the canister **8** (the outer peripheral surface of the canister shell **81**). The annular space **79** is provided appropriately along the entire length of the canister **8** in the up-down direction.

(16) FIG. **2** is a diagram illustrating an area in the vicinity of the upper end portion **86** of the canister **8**. As illustrated in FIG. **2**, the inside diameter of the upper end portion of the cask body **71** is greater than the inside diameter thereof at the other positions such as the central portion in the up-down direction. In other words, the inner peripheral surface of the upper end portion of the cask body **71** has a ring-shaped notch **713** extending along the entire perimeter of the cask body **71** (the entire perimeter in a circumferential direction having the central axis of the cask body **71** as its center). The ring-shaped notch **713** faces the upper end portion **86** of the canister **8** in a radial direction having the central axis of the cask body **71** as its center. Thus, the width of the annular space **79** around the upper end portion **86** of the canister **8** is greater than the width of the annular space **79** around the other portions of the canister **8** such as the central portion.

(17) The ring-shaped weir **12** has a cylindrical shape and, in the example illustrated in FIG. **2**, has an approximately cylindrical shape like the cask body **71**. The ring-shaped weir **12** is attached to the upper portion of the cask body **71** and projects upward from this upper portion. Preferably, the ring-shaped weir **12** may be fixedly attached to the upper end face of the cask body **71** by bolting using bolt holes that are used to fixedly attach the cask lid **73** (see FIG. **8**) to the cask body **71**. The bottom face of the ring-shaped weir **12** is provided with a ring-shaped seal member (not shown) in order to disable the passage of fluid between the bottom face of the ring-shaped weir **12** and the upper end face of the cask body **71**. Alternatively, the ring-shaped weir **12** may be attached to a face other than the upper end face in the upper portion of the cask body **71**. In the example illustrated in FIG. **2**, the upper end face of the cask body **71** is located at approximately the same level as the upper face of the canister lid **83** (i.e., the upper end face of the canister **8**) or slightly above the upper face of the canister lid **83**. The upper end of the ring-shaped weir **12** is located sufficiently above the upper face of the canister lid **83**.

(18) As illustrated in FIG. **1**, the water filler **2** includes an outflow line **21**, a clean water tank **22**, a heat exchanger **23**, an inflow line **24**, and a conveying pump **25**. The outflow line **21** connects the heat exchanger **23** and the upper water hole **711** of the cask body **71**. The heat exchanger **23** will be described later. The outflow line **21** is provided with a valve. The clean water tank **22** stores clean water, i.e., water that contains no radioactive materials. The clean water may, for example, be deionized water. The clean water tank **22** is connected to the outflow line **21**. The inflow line **24** connects the heat exchanger **23** and the lower water hole **712** of the cask body **71**. The inflow line **24** is provided with the conveying pump **25** and a valve.

(19) The water filler **2** is capable of driving the conveying pump **25** and supplying the clean water stored in the clean water tank **22** to the annular space **79** of the cask body **71** via the outflow line **21**, the heat exchanger **23**, the inflow line **24**, and the lower water hole **712**. In the state illustrated

in FIG. 1, the aforementioned annular space **79** is filled with the clean water. The space above the upper end portion **86** of the canister **8** is also filled with the clean water, so that the surface of the clean water is formed in the ring-shaped weir **12**. The ring-shaped weir **12** is provided with an overflow tube **13**, and when the surface of the clean water in the ring-shaped weir **12** becomes higher than a discharge port of the overflow tube **13**, the clean water is discharged via the overflow tube **13** to the clean water tank **22**. In FIG. 1, the spaces filled with the clean water are hatched with broken lines, and the pooled water in the storage pool P is not hatched in order to distinguish between the clean water that contains no radioactive materials and the pooled water that contains radioactive materials (the same applies to the other drawings). In the following description, the water that fills the interiors of the cask body **71** and the ring-shaped weir **12** (i.e., the water that fills the annular space **79** and the space above the upper end portion **86** of the canister **8**) is referred to as “fill water.”

(20) With the annular space **79** filled with the clean water, the water filler **2** drives the conveying pump **25** so that the clean water (i.e., fill water) in the annular space **79** flows out through the upper water hole **711** to the outflow line **21** and conveyed to the heat exchanger **23**. The clean water that has passed through the heat exchanger **23** flows into (returns to) the annular space **79** through the inflow line **24** and the lower water hole **712**. In this way, the water filler **2** is capable of circulating the clean water along a circulation path that starts from the annular space **79**, passes through the upper water hole **711**, the outflow line **21**, the heat exchanger **23**, the inflow line **24**, and the lower water hole **712** in sequence, and returns to the annular space **79**.

(21) The heat exchanger **23** includes a heat exchange part **231** and a pooled water circulator **232**. The pooled water circulator **232** includes a pump **233** and circulates the pooled water between the storage pool P and the heat exchange part **231**. The heat exchange part **231** exchanges heat between the clean water that flows in from the outflow line **21** and the pooled water that is circulated by the pooled water circulator **232**. In the present embodiment, since the fuel assemblies **9** in the canister **8** have high heat values, the heat exchanger **23** cools the clean water flowing in from the outflow line **21** with the pooled water. When the fuel assemblies **9** have low heat values, the cooling by the heat exchanger **23** (heat exchange between the clean water in the annular space **79** and the pooled water) may be omitted.

(22) As illustrated in FIGS. 1 and 2, the compressive stress applicator **3** includes a nozzle **31**, a water supplier **32**, a nozzle mover **33**, and a supporter **34**. The nozzle **31** is disposed in the fill water inside the cask body **71** and the ring-shaped weir **12**. The water supplier **32** illustrated in FIG. 1 includes a supply line **321**, a pump **322**, a supply tank **323**, and a filter **324**. One end of the supply line **321** is connected to the nozzle **31**, and the other end thereof is connected to the inflow line **24**. In the supply line **321**, the filter **324**, the supply tank **323**, and the pump **322** are arranged in the order specified from the inflow line **24** toward the nozzle **31**. Part of the clean water flowing in the inflow line **24** flows into the supply tank **323** after unnecessary substances are removed by the filter **324**. The supply tank **323** stores the clean water. The pump **322** increases the pressure of the clean water in the supply tank **323** and supplies the clean water to the nozzle **31**. Accordingly, high pressure water is sprayed from the nozzle **31** in the fill water.

(23) As illustrated in FIG. 2, the supporter **34** is fixedly attached to the upper face of the canister lid **83**. For example, the upper face of the canister lid **83** may be fixedly attached to a shield plate **51**, and the supporter **34** may be attached to the shield plate **51**. For example, the shield plate **51** may have an approximately disk-like shape and may be formed of a material that shields radioactive rays. The nozzle mover **33** includes a first movement mechanism **331**, a second movement mechanism **332**, and a third movement mechanism **333**. The first movement mechanism **331** moves the nozzle **31** in the up-down direction. The second movement mechanism **332** moves the nozzle **31** together with the first movement mechanism **331** in the radial direction. The third movement mechanism **333** is supported by the supporter **34** and moves (turns) the nozzle **31** together with the first and second movement mechanisms **331** and **332** in the circumferential direction.

(24) FIG. 3 is a flowchart of processing relating to the loading of the fuel assemblies **9** into the canister **8**. First, as illustrated in FIG. 4, the cask body **71** is placed on the cask holder **11** on the operation floor **F**, and the canister shell **81** is disposed in the cask body **71** (step **S11**). The cask body **71** has a lower opening that is closed by the cask bottom **72** and an upper opening that is open. The canister shell **81** has a lower opening that is closed by the canister bottom **82** and an upper opening that is open.

(25) Moreover, a ring-shaped seal member **52** (e.g., inflatable seal) is provided between the inner peripheral surface of the cask body **71** and the outer peripheral surface of the canister shell **81**. For example, the seal member **52** may be arranged in the vicinity of the underside of the ring-shaped notch **713**. This disables the passage of fluid between the seal member **52** and the inner peripheral surface of the cask body **71** and between the seal member **52** and the outer peripheral surface of the canister shell **81**. The cask body **71** is also connected to the water filler **2** described previously. When the water filler **2** drives the conveying pump **25**, the portion of the annular space **79** in the cask body **71** which is located below the seal member **52** is filled with the clean water in the clean water tank **22**.

(26) Then, as illustrated in FIG. 5, the cask body **71** is immersed together with the canister shell **81** in the storage pool **P**, using an overhead travelling crane or the like provided in a nuclear reactor building. For example, the cask body **71** may be disposed on a cask holder **61** provided at the bottom of the storage pool **P**. For convenience of illustration, the positions of the upper water hole **711**, the lower water hole **712**, and the water filler **2** relative to the cask body **71** in FIG. 5 are reversed right to left from those in FIG. 4. Thereafter, the fuel assemblies **9** stored in the storage pool **P** are transported (inserted) into the canister shell **81** and loaded therein by a fuel exchanging machine **62** (step **S12**).

(27) When a predetermined number of fuel assemblies **9** has been loaded in the canister shell **81**, the canister lid **83** is fitted in the upper opening of the canister shell **81** in the storage pool **P** (see FIG. 6). Thereafter, the cask body **71** is taken out of the storage pool **P**, together with the canister shell **81** (step **S13**). In the water filler **2**, the clean water is continuously circulated between the annular space **79** and the heat exchanger **23** so as to reduce the possibility that the temperature inside the canister shell **81** may be excessively increased outside the storage pool **P** by the heat generated by the fuel assemblies **9** in the canister shell **81**.

(28) The cask body **71** taken out of the storage pool **P** may be subjected to, for example, decontamination of the outer surface in a decontamination pit and thereafter placed on the cask holder **11** on the operation floor **F** as illustrated in FIG. 6. In FIG. 6, the canister **8** filled with the pooled water is hatched with a different pattern from that of the annular space **79** filled with the clean water. Then, the pooled water that is accumulated above the canister lid **83** and in a portion of the annular space **79** located above the seal member **52** is removed by a drainage system **63** that includes a pump **631**. Moreover, for example, the upper face of the canister lid **83** may be decontaminated as necessary. The removal of the pooled water and the decontamination of, for example, the canister lid **83** described above may be performed in a decontamination pit.

(29) Moreover, the shield plate **51** described previously is fixedly attached to the upper face of the canister lid **83**, and the ring-shaped shield **53** is fitted in the ring-shaped notch **713**. Like the shield plate **51**, the ring-shaped shield **53** is formed of a material that shields radioactive rays. The outside diameter of the ring-shaped shield **53** is slightly smaller than the inside diameter of the cask body **71** at the position of the ring-shaped notch **713**. For example, the inside diameter of the ring-shaped shield **53** may be approximately the same as or slightly smaller than the inside diameter of the cask body **71** at the positions other than the ring-shaped notch **713**. In the state illustrated in FIG. 6, radioactive rays (e.g., neutrons) emitted from the fuel assemblies **9** are shielded by the clean water that fills the portion of the annular space **79** that is located below the seal member **52**. In the case where the ring-shaped shield **53** is attached to the ring-shaped notch **713**, the range of the canister shell **81** that can shield radioactive rays is equivalent to or more than that in the case where the

ring-shaped notch **713** is not provided.

(30) After the attachment of the shield plate **51** and the ring-shaped shield **53**, welding equipment **64** is attached to the canister lid **83**. For example, the welding equipment **64** illustrated in FIG. **6** may be laser welding equipment and include a welding head **641**, a head mover **642**, and a supporter **646**. The welding head **641** emits laser light. The head mover **642** and the supporter **646** are similar to the nozzle mover **33** and the supporter **34** of the compressive stress applicator **3** illustrated in FIG. **2**. The supporter **646** is fixedly attached to the canister lid **83** via the shield plate **51**. The head mover **642** includes a first movement mechanism **643**, a second movement mechanism **644**, and a third movement mechanism **645**. The first movement mechanism **643** moves the welding head **641** in the up-down direction. The second movement mechanism **644** moves the welding head **641** together with the first movement mechanism **643** in the radial direction. The third movement mechanism **645** is supported by the supporter **646** and moves (turns) the welding head **641** together with the first and second movement mechanisms **643** and **644** in the circumferential direction.

(31) In the welding equipment **64**, the welding head **641** that is emitting laser light is moved by the head mover **642**. Specifically, the position that is to be irradiated with the laser light emitted from the welding head **641** shifts along the outer peripheral edge of the canister lid **83**. Accordingly, the outer peripheral edge of the canister lid **83** and the edge of the upper opening of the canister shell **81** are welded together along the entire perimeter. That is, the canister lid **83** is welded to the upper opening of the canister shell **81** (step **S14**). Preferably, the canister **8** may be provided with a ring-shaped notch that extends along the outer peripheral edge of the upper face of the canister lid **83**, and a ring-shaped member may be fitted in this notch after the welding of the canister lid **83**. Then, the ring-shaped member is welded to the canister lid **83** and the canister shell **81** by the welding equipment **64**. The welding method used by the welding equipment **64** may be other than laser welding (e.g., arc welding).

(32) Note that part of the pooled water may be discharged from the inside of the canister **8** before the welding of the canister lid **83**. For example, the canister lid **83** may be provided with a plurality of ports **831** (see FIG. **7** described later), and these ports **831** may be used to discharge part of the pooled water and to supply an inert gas into the canister **8**. This reduces the influence of water vapor during welding. For example, the shield plate **51** may be provided with openable and closable holes at positions that face the ports **831** of the canister lid **83**, and these holes may be used to connect various tubes to the ports **831**. After the welding of the canister lid **83**, the soundness of the welds is preferably checked by conducting a predetermined test or inspection. The same applies to the welding of the ring-shaped member and the welding of port covers described later.

(33) When the welding of the canister lid **83** (and the ring-shaped member) is completed, the welding equipment **64** is dismantled from the canister lid **83**. Then, all the pooled water in the canister **8** is discharged through the ports **831**. After the discharge of the pooled water, connecting tubes of a vacuum drier **65** are connected to the ports **831** of the canister lid **83** as illustrated in FIG. **7**, and vacuum drying is performed in the canister **8**. When the vacuum drying in the canister **8** is completed, gas supply tubes are connected to the ports **831**, and a predetermined gas (e.g., helium (He) gas) is charged into the canister **8** (step **S15**). Preferably, port covers may be welded to the ports **831** of the canister **8** after charging of the gas. In this case, the welding equipment **64** is dismantled from the canister lid **83** after the welding of the port covers is completed.

(34) Then, as illustrated in FIG. **1**, the residual stress improvement apparatus **1** is installed with respect to the cask body **71** and the canister **8** (step **S16**). At this time, the water filler **2** has already been attached to the cask body **71** (before the cask body **71** is immersed in the storage pool **P**) as illustrated in FIG. **7**. Thus, the installation of the residual stress improvement apparatus **1** includes attaching the ring-shaped weir **12**, the overflow tube **13**, and the compressive stress applicator **3** to the cask body **71** and the canister **8**.

(35) Specifically, the ring-shaped shield **53** is dismounted, and the ring-shaped weir **12** is attached to the upper portion of the cask body **71**. Moreover, both ends of the overflow tube **13** are connected respectively to the ring-shaped weir **12** and the clean water tank **22**. In the compressive stress applicator **3**, the supporter **34** that supports the nozzle **31** and the nozzle mover **33** is fixedly attached to the canister lid **83** via the shield plate **51**. One end of the supply line **321** of the water supplier **32** (the end on the side opposite the nozzle **31**) is connected to the inflow line **24**. Through the operations described above, the installation of the residual stress improvement apparatus **1** is completed. Alternatively, the shield plate **51** may be dismounted from the canister lid **83**, and the supporter **34** may be fixedly attached directly to the canister lid **83**.

(36) In the present example of processing, after the removal of the seal member **52** between the inner peripheral surface of the cask body **71** and the outer peripheral surface of the canister **8**, the water filler **2** further supplies the clean water stored in the clean water tank **22** to the interior of the cask body **71** via the inflow line **24**. Accordingly, the portion of the annular space **79** that is located above the position of the seal member **52** is also filled with the clean water, and the surface of the clean water is formed in the vicinity of the discharge port of the overflow tube **13** in the ring-shaped weir **12** as illustrated in FIG. **1**. That is, the space above and around the upper end portion **86** of the canister **8** is also filled with the clean water in the cask body **71** and the ring-shaped weir **12** (step S17).

(37) The clean water that fills the space above and around the upper end portion **86** of the canister **8** shields radioactive rays that are emitted from the fuel assemblies **9** and coming upward.

(38) In order to allow the layer of the clean water that is located above the upper end portion **86** to have a certain degree of capability to shield the radioactive rays, the layer of the clean water has a thickness (i.e., distance from the upper face of the canister lid **83** to the surface of the clean water) of, for example, 10 cm or more, preferably 30 cm or more, and more preferably 50 cm or more.

(39) In the compressive stress applicator **3**, the nozzle **31** is disposed in the clean water in the ring-shaped weir **12** and the cask body **71**, i.e., in the fill water. For example, the nozzle **31** may have a spray outlet that faces the outer surface (the outer peripheral surface or the upper face) of the upper end portion **86** of the canister **8**. When the pump **322** of the water supplier **32** is driven, high pressure water is sprayed from the nozzle **31** toward the outer surface of the upper end portion **86** in the fill water. Accordingly, cavitation occurs in the fill water, and impact pressure that occurs upon collapse of air bubbles causes plastic deformation of a region of the outer surface of the upper end portion **86** that approximately faces the spray outlet of the nozzle **31**. That is, compressive stress is applied to this region of the canister **8** by water-jet peening.

(40) In the compressive stress applicator **3**, the nozzle **31** that is spraying high pressure water is moved by the nozzle mover **33**. Specifically, the nozzle **31** sprays high pressure water in succession to approximately the entire region of the outer peripheral surface of the upper end portion **86** of the canister **8** and to the region of the upper face of the upper end portion **86** that is located outside the vicinity of the outer peripheral edge of the canister lid **83** (hereinafter, these regions are collectively referred to as “target regions”). In FIG. **2** that illustrates a cross section of the canister **8**, the target regions of the outer surface of the upper end portion **86** are indicated by thick solid lines affixed with reference signs A1. The entire target regions are located in the fill water. In this way, compressive stress is applied to the target regions of the upper end portion **86** of the canister **8**, and residual tensile stress caused by the welding is reduced (step S18). Steps S17 and S18 described above correspond to residual stress improvement processing for improving residual stress in the canister **8**. In the residual stress improvement processing, it is preferable that residual stress is shifted from the tensile side to the compressive side.

(41) As described previously, the presence of the ring-shaped notch **713** of the cask body **71** allows the width of the annular space **79** around the upper end portion **86** of the canister **8** to become greater than the width of the annular space **79** around the other portions of the canister **8** such as the central portion. This makes it possible to dispose the nozzle **31** at the position facing the outer

peripheral surface of the upper end portion **86** and to readily apply compressive stress to the outer peripheral surface. The residual stress improvement apparatus **1** may include a plurality of nozzles **31** in order to complete the application of compressive stress to the target regions in a short time. The target regions to which compressive stress is applied may be changed appropriately.

(42) While the compressive stress applicator **3** applies compressive stress to the upper end portion **86**, the water filler **2** in FIG. **1** continuously circulates the clean water along the circulation path that starts from the annular space **79**, passes in sequence through the upper water hole **711**, the outflow line **21**, the heat exchanger **23**, the inflow line **24**, and the lower water hole **712**, and returns to the annular space **79**. This reduces the possibility that the temperature in the canister **8** may be increased excessively. Alternatively, the valves of the outflow line **21** and the inflow line **24** may be closed so that the clean water is circuited along a circulation path that starts from the interior of the cask body **71** (and the interior of the ring-shaped weir **12**), passes in sequence through the overflow tube **13**, the clean water tank **22**, the outflow line **21**, the heat exchanger **23**, the inflow line **24**, the supply line **321**, and the nozzle **31**, and returns to the interior of the cask body **71** (while high pressure water is being sprayed from the nozzle **31**).

(43) Since the supply line **321** of the water supplier **32** is connected to the flow path of the clean water flowing out of the annular space **79** (in FIG. **1**, the inflow line **24**), the clean water whose temperature is increased by the heat generated by the fuel assemblies **9** in the canister **8** is supplied to the nozzle **31**. During water-jet peening, cavitation is likely to occur at temperatures higher than ordinary temperature (e.g., 40 to 60° C.). Therefore, the compressive stress applicator **3** that sprays the water heated in the annular space **79** from the nozzle **31** becomes capable of applying compressive stress to the upper end portion **86** of the canister **8** more efficiently than in the case of using ordinary-temperature water. Depending on the heat values of the fuel assemblies **9** or other features, the supply line **321** of the water supplier **32** may be connected to the outflow line **21** so as to allow the clean water before passing through the heat exchange part **231** to flow into the supply line **321**. As another alternative, the supply tank **323** may include a heat regulator that controls the temperature of the clean water.

(44) When the application of compressive stress to the upper end portion **86** of the canister **8** is completed, the water filler **2** stops to drive the pump **25**. Also, the clean water in the interiors of the cask body **71** and the ring-shaped weir **12** is discharged through, for example, the lower water hole **712**. Alternatively, the cask bottom **72** may have an openable and closable opening, and the clean water in the interiors may be discharged through this opening. When the discharge of the clean water in the interiors is completed, the water filler **2**, the ring-shaped weir **12**, the overflow tube **13**, and the compressive stress applicator **3** are dismounted from the cask body **71** and the canister **8**. The shield plate **51** is also removed from the canister lid **83**.

(45) Then, as illustrated in FIG. **8**, the cask lid **73** is fixedly attached to the upper opening of the cask body **71** by, for example, bolting so as to close the upper opening (step **S19**). This configures the cask **7** that serves as a closed container that houses the canister **8**. In the example illustrated in FIG. **8**, connecting tubes of a vacuum driver **66** are connected to the ports **731** of the cask lid **73**, and vacuum drying is performed in the cask **7**. When the vacuum drying in the cask **7** is completed, gas supply tubes are connected to the ports **731** so that the cask **7** is filled with a predetermined gas (e.g., helium (He) gas). The vacuum drying and the application of the gas in the cask **7** may be omitted. Through the operations described above, the processing relating to the loading of the fuel assemblies **9** into the canister **8** is completed. The canister **8** that is housed in the cask **7** is transported to the storage facility.

(46) As described above, in the method of improving residual stress in the canister **8**, when the canister **8** having the canister shell **81** and the canister lid **83** welded to the upper opening of the canister shell **81** is disposed in the cask body **71** (see FIG. **1**), the space above and around the upper end portion **86** of the canister **8** in the cask body **71** is filled with fill water. Then, water is sprayed from the nozzle **31** disposed in the fill water to apply compressive stress by water-jet peening to a

predetermined region of the upper end portion **86** of the canister **8**. This makes it possible to readily apply compressive stress to the canister **8** while shielding radioactive rays by the fill water. As a result, it is possible to prevent or suppress stress-corrosion cracking in the canister **8**. The water-jet peening can also suppress the generation of unnecessary substances during the application of compressive stress.

(47) In the residual stress improvement method, the ring-shaped weir **12** is attached to the upper portion of the cask body **71** to project upward therefrom, and the surface of water is formed in the ring-shaped weir **12**. This makes it possible to readily fill the space above the upper end portion **86** of the canister **8** with water. Besides, it is possible to readily increase the thickness of the layer of water that fills the space above the upper end portion **86** and thereby to further improve the capability to shield radioactive rays that are emitted from the fuel assemblies **9** and coming upward. In the case where the upper end of the cask body **71** is located to some extent above the upper face of the canister lid **83**, the ring-shaped weir **12** may be omitted.

(48) The annular space **79** is formed between the inner peripheral surface of the cask body **71** and the outer peripheral surface of the canister **8**, and the annular space **79** is filled with water. Thus, the water that fills the annular space **79** can also be used to shield the radioactive rays emitted from the fuel assemblies **9**. In the application of compressive stress to the upper end portion **86** of the canister **8**, the water sprayed from the nozzle **31** includes the water that has filled the annular space **79**. Accordingly, it is possible to spray water with a favorable temperature from the nozzle **31** using the heat generated by the fuel assemblies **9** and to improve the efficiency of applying compressive stress. Depending on the design, the water supplier **32** may be provided independently of the circulation path of water including the annular space **79**, and water that does not pass through the annular space **79** may be supplied to the nozzle **31**.

(49) Preferably, the width of the annular space **79** around the upper end portion **86** of the canister **8** is greater than the width of the annular space **79** around the central portion of the canister **8**. In this way, the presence of the space in which the nozzle **31** can be disposed between the outer peripheral surface of the upper end portion **86** and the inner peripheral surface of the cask body **71** makes it possible to readily apply compressive stress to the outer peripheral surface of the upper end portion **86**. Moreover, when the canister lid **83** is welded to the upper opening of the canister shell **81**, the ring-shaped shield **53** is provided around the upper end portion **86** of the canister **8** (see FIG. 6). As a result, it is possible to suppress leakage of radioactive rays from the aforementioned space between the upper end portion **86** and the cask body **71** during the welding of the canister lid **83**. Depending on the design of the cask **7**, the ring-shaped notch **713** may be omitted, and the width of the annular space **79** may be made constant along the entire length in the up-down direction.

(50) The residual stress improvement method and the residual stress improvement apparatus **1** described above may be modified in various ways.

(51) Although in the embodiment described above, water-jet peening is used to apply compressive stress to the upper end portion **86** of the canister **8**, any other technique may be used to apply compressive stress to the upper end portion **86** in the fill water. For example, in the case where laser peening is adopted, laser light is emitted from a head disposed in the fill water toward the upper end portion **86** of the canister **8**. In this case, it is also possible to suppress the generation of unnecessary substances during the application of compressive stress as in the case of using water-jet peening. If the contamination of water around the upper end portion **86** does not become a problem, other techniques may be adopted, such as banishing using a ball or a roller.

(52) The annular space **79** formed between the inner peripheral surface of the cask body **71** and the outer peripheral surface of the canister **8** may be filled with water (e.g., pooled water) other than clean water. The same applies to the water that fills the space above and around the upper end portion **86** of the canister **8** in step **S17** in FIG. 3 and the water that is sprayed from the nozzle **31** in step **S18**. On the other hand, in order to save work of decontaminating the inner peripheral surface of the cask body **71** and the outer surface of the canister **8**, it is preferable that the annular space **79**

is filled with clean water that contains no radioactive materials as in the embodiment described above. Similarly, it is preferable that the water that fills the space above and around the upper end portion **86** of the canister **8** in step **S17** and the water sprayed from the nozzle **31** in step **S18** may be clean water.

(53) In the example of processing described above, when the space above and around the upper end portion **86** of the canister **8** is filled with clean water, the seal member **52** provided between the inner peripheral surface of the cask body **71** and the outer peripheral surface of the canister **8** is removed. However, in the presence of the seal member **52**, the space above and around the upper end portion **86** of the canister **8** may be filled with clean water, using, for example, the nozzle **31**. As described previously, in the case where the fuel assemblies **9** have low heat values, the portion of the annular space **79** that is located below the seal member **52** does not necessarily have to be filled with water, and in this case, only the space above the seal member **52**, i.e., the space above and around the upper end portion **86** of the canister **8**, may be filled with clean water in step **S17**. In the case where the aforementioned portion of the annular space **79** is not filled with water, it is preferable that the pressure in this portion is set to be higher than or equal to a predetermined value. Moreover, the gas (e.g., air) that fills the aforementioned portion may be cooled by heat exchange performed by the heat exchanger **23**.

(54) The configurations of the above-described preferred embodiments and variations may be appropriately combined as long as there are no mutual inconsistencies.

(55) While the invention has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore to be understood that numerous modifications and variations can be devised without departing from the scope of the invention.

REFERENCE SIGNS LIST

(56) **1** Residual stress improvement apparatus **2** Water filler **3** Compressive stress applicator **8** Canister **9** Fuel assembly **12** Ring-shaped weir **31** Nozzle **53** Ring-shaped shield **71** Cask body **79** Annular space **81** Canister shell **83** Canister lid **86** Upper end portion of (canister) **S11 to S19** Step

Claims

1. A residual stress improvement method of improving residual stress in a canister in which a fuel assembly is loaded, the residual stress improvement method comprising: a) disposing a cylindrical canister shell in a cylindrical cask body to form an annular space between an inner peripheral surface of said cask body and an outer peripheral surface of the canister shell, and providing a ring-shaped seal member in said annular space, said canister shell having a lower opening that is closed, said cask body having a lower opening that is closed; b) filling with clean water a portion of said annular space that is located below said seal member; c) immersing said cask body together with said canister shell in a storage pool to load the fuel assembly in said canister shell; d) fitting a canister lid in an upper opening of said canister shell in said storage pool, and taking said cask body out of said storage pool together with said canister shell; e) decontaminating said cask body; f) welding said canister lid to said upper opening of said canister shell; g) filling with fill water a space above and around an upper end portion of said canister in said cask body, said canister including said canister shell and said canister lid; and h) applying compressive stress to a predetermined region of said upper end portion of said canister in the fill water.

2. The residual stress improvement method according to claim 1, wherein in said operation h), water-jet peening is used to apply compressive stress to said predetermined region, the water-jet peening being performed by injecting water from a nozzle that is disposed in the fill water.

3. The residual stress improvement method according to claim 2, wherein in said operation h), said seal member is removed from said annular space, and the water injected from said nozzle includes the water that has filled said annular space and whose temperature has been increased by heat generated by said fuel assembly in said canister.

4. The residual stress improvement method according to claim 2, wherein the fill water in said operation g) and the water injected from said nozzle in said operation h) are clean water that contains no radioactive materials.
 5. The residual stress improvement method according to claim 1, wherein in said operation g), a surface of water is formed in a ring-shaped weir that is attached to an upper portion of said cask body and projects upward therefrom.
 6. The residual stress improvement method according to claim 1, wherein a width of said annular space around said upper end portion of said canister is greater than a width of said annular space around a central portion of said canister, and wherein, in said operation f), when said canister lid is welded to said upper opening of said canister shell, a ring-shaped shield is placed around said upper end portion of said canister.
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